

Computer Organization & Architecture

Understand how CPU, memory, data paths, control units, I/O and instruction execution fit together.

Recommended learning order

1. Number systems and data representation
2. Memory hierarchy and interfacing
3. Instruction sets and addressing modes
4. ALU, data path and control unit
5. I/O and pipelining

Direct YouTube resources

Source	What to use it for	Open
Neso Academy	Computer Organization & Architecture playlist.	YouTube →
Neso Academy	Computer Organization & Architecture course page with chapters.	YouTube →
Gate Smashers	Computer Organization and Architecture playlist.	YouTube →

Tip: open the YouTube → buttons directly from the PDF. This roadmap is based on the supplied Hardware Learning Roadmap, with Computer Networks added as a separate course because you specifically requested it.

Prepared as a separate topic-wise learning PDF.